

Finals Task 3. Using Time Intelligence Function

Description:

This lab focused on using DAX Time Intelligence functions in Power BI to analyze sales performance over time. The activity involved creating measures that manipulate date-based filter contexts to generate meaningful business insights.

A Year-to-Date (YTD) sales measure was first created to track accumulated sales values throughout each year. The measure was then added to a matrix visualization to observe how sales values increase over time across months and years.

The lab also introduced the use of variables in DAX by creating a Year-over-Year (YoY) Growth measure. This calculation compared sales performance between different years to identify growth trends and percentage changes. The results were formatted and validated using matrix visuals to ensure accurate calculations.

Through this activity, skills in DAX formulas, time intelligence functions, variable usage, and analytical reporting in Power BI were strengthened.

The screenshot displays the Power BI Desktop interface. The main area shows a matrix visualization of sales data. The columns are: Year, Avg Price, Median Price, Min Price, Max Price, Orders, Order Lines, Sum of Sales, Sales YTD, and Sales YoY Growth. The rows are grouped by fiscal year (FY2018, FY2019, FY2020) and then by month. The 'Sales YTD' column shows the cumulative sales for each year, and the 'Sales YoY Growth' column shows the percentage change from the previous year. The 'Sales YoY Growth' column is formatted as a percentage.

Year	Avg Price	Median Price	Min Price	Max Price	Orders	Order Lines	Sum of Sales	Sales YTD	Sales YoY Growth
FY2018	\$748.68	\$419.46	\$4.75	\$2,146.96	739	8,459	\$16,429,043	P16,429,042.6	
2017 Jul	\$655.59	\$419.46	\$5.19	\$2,146.96	38	352	\$489,328.4	P489,328.4	
2017 Aug	\$758.93	\$419.46	\$4.75	\$2,146.96	75	785	\$1,540,072	P2,029,400.42	
2017 Sep	\$748.85	\$419.46	\$5.19	\$2,146.96	60	593	\$1,166,332	P3,195,732.87	
2017 Oct	\$677.45	\$419.46	\$5.19	\$2,146.96	40	499	\$844,833	P4,040,565.58	
2017 Nov	\$752.31	\$419.46	\$5.01	\$2,146.96	90	1,106	\$2,325,755	P6,366,320.49	
2017 Dec	\$734.58	\$419.46	\$5.01	\$2,146.96	63	803	\$1,703,435	P8,069,755.58	
2018 Jan	\$808.94	\$419.46	\$5.19	\$2,146.96	40	377	\$713,230	P8,782,985.12	
2018 Feb	\$96.80	\$419.46	\$5.01	\$2,146.96	79	866	\$1,900,794	P10,683,779.58	
2018 Mar	\$863.54	\$419.46	\$5.19	\$2,146.96	64	653	\$1,455,280	P12,139,059.85	
2018 Apr	\$732.25	\$419.46	\$5.19	\$2,146.96	37	494	\$880,011	P13,022,070.83	
2018 May	\$761.30	\$419.46	\$4.75	\$2,146.96	85	1,112	\$2,269,720	P15,291,790.7	
2018 Jun	\$552.95	\$419.46	\$4.75	\$2,146.96	68	819	\$1,137,252	P16,429,042.6	
FY2019	\$397.81	\$202.33	\$4.32	\$1,466.01	1,255	21,670	\$27,979,780	P27,979,779.53	70.31%
2018 Jul	\$350.74	\$196.33	\$4.75	\$1,466.01	72	1,723	\$2,411,559	P2,411,558.85	392.83%
2018 Aug	\$372.75	\$202.33	\$4.75	\$1,466.01	139	2,964	\$3,615,914	P6,027,473.31	134.79%
2018 Sep	\$365.30	\$196.33	\$4.32	\$1,466.01	111	2,185	\$2,894,647	P8,922,119.92	148.18%
2018 Oct	\$382.00	\$202.33	\$5.19	\$1,466.01	73	1,406	\$1,884,177	P10,726,296.85	113.55%
2018 Nov	\$412.48	\$209.26	\$5.01	\$1,466.01	133	2,345	\$3,054,997	P13,781,293.57	31.36%
2018 Dec	\$407.08	\$202.33	\$4.75	\$1,466.01	114	1,732	\$2,188,206	P15,969,499.23	28.46%
2019 Jan	\$428.87	\$224.45	\$5.19	\$1,466.01	65	983	\$1,318,592	P17,288,091.24	84.88%
2019 Feb	\$488.75	\$469.79	\$5.19	\$1,466.01	132	1,664	\$2,386,073	P19,674,164.43	25.53%
2019 Mar	\$450.03	\$324.45	\$5.01	\$1,466.01	106	1,215	\$1,564,295	P21,238,459.35	7.49%
2019 Apr	\$369.78	\$198.04	\$5.01	\$1,466.01	74	1,449	\$1,864,433	P23,106,892.7	111.60%
2019 May	\$400.51	\$202.33	\$4.75	\$1,466.01	124	2,306	\$2,882,638	P25,989,530.28	27.00%
2019 Jun	\$390.32	\$202.33	\$4.75	\$1,466.01	102	1,698	\$1,990,249	P27,979,779.53	75.01%
FY2020	\$392.12	\$200.05	\$1.33	\$1,466.01	1,622	27,722	\$33,139,748	P33,139,748.07	18.44%
2019 Jul	\$350.55	\$158.43	\$1.37	\$1,466.01	94	2,186	\$2,729,167	P2,729,167.03	13.17%
2019 Aug	\$342.19	\$153.89	\$1.33	\$1,466.01	185	3,754	\$4,306,549	P7,035,716.17	19.10%
2019 Sep	\$343.40	\$158.43	\$1.37	\$1,466.01	176	3,760	\$4,153,399	P11,189,115.52	43.49%
2019 Oct	\$378.65	\$158.43	\$1.37	\$1,466.01	99	1,810	\$2,293,200	P13,482,315.95	27.11%
2019 Nov	\$395.80	\$200.05	\$1.37	\$1,466.01	178	2,804	\$3,490,438	P16,972,753.98	14.25%
Total	\$446.39	\$214.24	\$1.33	\$2,146.96	3,616	57,851	\$77,548,570		0.00%